Marmara Lectures

MARMARA LECTURES

Review of Complex Numbers

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In CSE 348 we assume a basic knowledge of complex numbers. So what we will do here is just a review rather than a complete exposition of the subject. Student must refer to many excellent internet resources (esp. youtube videos) if he/she feels lacking in this area.

Basics

$$i = j = \sqrt{-1}$$

Electrical engineers use j and mathematicians use i when describing $\sqrt{-1}$. We will go with mathematicians and use i, but be warned that your textbook (Oppenheim-Willsky) uses electrical engineering convention j.

By definition

$$i.\, i = i^2 = \sqrt{-1}.\, \sqrt{-1} = (\sqrt{-1})^2 = -1$$

An imaginary number is a real number multiplied by the imaginary unit i. Examples are $5i$, $-2.3i$, $\pi^2 i$ etc. Hence the general imaginary number can be expressed as ai, where $a \in \mathbb{R}$. The square of the imaginary number ai is $-a^2$.

A complex number z has a real part and a imaginary part, $z = a + bi$, where $a, b \in \mathbb{R}$ Examples are $3 + 2i$, $-5.2 - 3.4i$, $e - \pi i$. Note that the letter z is usually used to denote the complex quantities.

Two associated functions with complex numbers are $\mathrm{Re}(.)$ and $\mathrm{Im}(.)$. One gives the real part and the other gives the imaginary part of a complex number

$$\mathrm{Re}(z) = \mathrm{Re}(a + ib) = a$$
$$\mathrm{Im}(z) = \mathrm{Im}(a + ib) = b$$

Examples are $\mathrm{Re}(3+7i)=3$ and $\mathrm{Re}(5-7.5i)=-7.5$

Conjugate (Turkish: Eşlenik) of the complex number $z=a+bi$ is

$$z^*=\overline{z}=(a+ib)^*=\overline{a+ib}=a-ib$$

Addition, subtraction, multiplication and division of complex numbers is always a complex number.In other words, result of these elementary operations can always be put into the form (some real number)+(some real number)i form.

Addition, subtraction and multiplication are done in the usual way. If $z_1=a+ib$, $z_2=c+id$, then

$$\begin{aligned} z_1+z_2&=a+c+(b+d)i\\ z_1-z_2&=a-c+(b-d)i\\ z_1.\,z_2&=(a+bi)(c+di)=ac+adi+cbi+bdi^2=ac-bd+(ad+cb)i \end{aligned}$$

Note that multiplication of a complex number by its conjugate always yields a real number:

$$z_1.\,\overline{z_1}=(a+bi)(a-bi)(a-bi)=a.\,a–a.\,bi–bi.\,a+bi.\,bi=a^2+b^2$$

Division is a bit more difficult.

$$\frac{z_1}{z_2}=\frac{a+ib}{c+id}$$

The RHS above is not in complex number form. To put it into a complex number form, we multiply the numerator and denominator by the conjugate of the denominator, as this will make the denominator a real number:

$$\frac{z_1}{z_2}=\frac{a+ib}{c+id}=\frac{(a+ib)(c-id)}{(c+id)(c-id)}=\frac{(ac+bd)+(bc-ad)i}{c^2+d^2}=\frac{ac+bd}{c^2+d^2}+\frac{bc-ad}{c^2+d^2}$$

Problem: What is $\mathrm{Re}\left(\frac{z_1}{z_2}\right)$ and $\mathrm{Im}\left(\frac{z_1}{z_2}\right)$?

Complex plane (or Argand plane)

Recall that there is a unique correspondance between the real numbers and a line, ie, each real number can be made to correspond to a point on a line. The result is called a “real line”.

A similar procedure can be done for complex numbers. Each complex number $z = a + ib$ can be put into a one-to-one correspondance to a unique point P in a plane if

- $\mathrm{Re}(z) = a$ is taken as the x-coordinate of P.
- $\mathrm{Im}(z) = b$ is taken as the y-coordinate of P.

The result is called a "complex plane" or an "Argand plane". Note that in argand plane adding/subtracting two complex numbers has the same meaning as adding/subtracting two vectors (ie, parallelogram rule). But do multiplication/division also has a similar meaning? We will see shortly.

Remember from your previous math lectures that a point in a plane can be given in two different coordinate systems: cartesian and polar. Describing a complex number $z = a + ib$ in the complex plane with the point (a, b) obviously corresponds to the carteisan case. Is it possible to describe the same point z using the polar coordinates? The answer is yes. The same point $z = a + ib$ can be described as $z = r\angle\theta$. From elementary geometry, we can compute the relationships between (a, b) and (r, θ):

$$\begin{aligned} r &= \sqrt{a^2 + b^2} = \sqrt{z\bar{z}} \\ \theta &= \arctan\frac{b}{a} \\ a &= r\cos(\theta) \\ b &= r\sin(\theta) \end{aligned}$$

Note that while the cartesian expression is unique, the polar expression is not unique: If we represent a point P with $r\angle\theta$, we can also represent it with $r\angle\theta + 2k\pi$, where $k \in \mathbb{I}$. As there are infinitely many integers, there are infinitely many equivalent polar representations of the same complex number. This will have very important consequences, which we will encounter very shortly. But, for a spoiler, just remember that each number has two square roots. For example, 4 has square roots 2 and -2. This is one such consequence..

Euler's formula

In 1740, Leonhard Euler discovered a very surprising representation for complex numbers. Given the polar representation (r, θ) of a complex number z, its cartesian representation can be written as

$$z = r\cos(\theta) + ir\sin(\theta)$$

Let us open $\cos(\theta)$ and $\sin(\theta)$ to their taylor series (you have to KNOW this step from your previous math lectures)

$$z = r\left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots\right) + ir\left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots\right)$$

using the properties of i, we get

$$\begin{aligned} z &= r\left(1 + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^6}{6!} + \dots\right) + r\left((i\theta) + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^5}{5!} + \frac{(i\theta)^7}{7!} + .\right. \\ &= r\left(1 + (i\theta) + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \frac{(i\theta)^6}{6!} + \frac{(i\theta)^7}{7!} + \dots\right) \\ &= re^{i\theta} \end{aligned}$$

We have proved that

$$re^{i\theta} = r\cos(\theta) + ir\sin(\theta)$$

or

$$e^{i\theta} = \cos(\theta) + ir\sin(\theta)$$

which is known as the Euler's formula.

Examples:

$$\begin{aligned} e^{i\frac{\pi}{2}} &= -1 \\ e^{i\pi} &= -1 \\ e^{i\frac{3\pi}{2}} &= -i \\ e^{i2\pi} &= 1 \\ e^{i\frac{\pi}{4}} &= \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \end{aligned}$$

With the aid of euler's formula, we can compute quantities which are thought to be uncomputable before. In this way, Euler's formula vastly expands our computational abilities.

A very important thing to note is that euler representation (like its origin, polar representation) is not unique.

$$e^{i\theta} = e^{i\theta + 2k\pi} \qquad k \in \mathbb{I}$$

This can be easily proveb as $e^{i\theta+2k\pi} = e^{i\theta}e^{i2k\pi} = e^{i\theta}$ as $e^{i2k\pi} = 1$ for $k \in \mathbb{I}$

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